

Lab07

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1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
path <- "data"
fname <- file.path(path, "database.sqlite3")
fname
```

```
[1] "data/database.sqlite3"
```

```
file.exists(fname)
```

```
[1] TRUE
```

```
library(DBI)
library(RSQLite)

conn <- dbConnect(SQLite(), dbname = fname)
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

```
SELECT DISTINCT i.id AS instructor_id, i.name AS professor
      FROM subjects AS su
      JOIN subject_memberships AS sm ON su.code = sm.subject_code
      JOIN sections AS se ON sm.course_offering_uuid = se.course_offering_uuid
      JOIN teachings AS t ON se.uuid = t.section_uuid
      JOIN instructors AS i ON t.instructor_id = i.id
      WHERE su.abbreviation = 'STAT'
      ORDER BY professor
```

Table 1: Displaying records 1 - 10

instructor_id	professor
4787911	ABIGAIL BENZINE
3111842	ADIN-CRISTIAN ANDREI
3775809	AKICHIKA OZEKI
4958796	ALAN HUANG
5451706	ALBERTO DEL PIA
3661782	ALBRECHT KLEMM
5656034	ALEXANDER COVINGTON
4680082	ALEXANDER FISH
3573775	ALEXANDER KISELEV
5162339	ALYSSA DIGILIO

```
SELECT COUNT() AS QUANTIDADE FROM (SELECT DISTINCT i.id AS instructor_id, i.name AS professor
FROM subjects AS su
JOIN subject_memberships AS sm ON su.code = sm.subject_code
JOIN sections AS se ON sm.course_offering_uuid = se.course_offering_uuid
JOIN teachings AS t ON se.uuid = t.section_uuid
JOIN instructors AS i ON t.instructor_id = i.id
WHERE su.abbreviation = 'STAT'
ORDER BY professor)
```

Table 2: 1 records

QUANTIDADE
379

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

- Quem é o professor mais difícil?

```
WITH stat_offerings AS (
SELECT sm.course_offering_uuid
FROM subject_memberships AS sm
JOIN subjects AS su ON sm.subject_code = su.code
WHERE su.abbreviation = 'STAT')
```

```

),
sec_gpa AS (
    SELECT gd.course_offering_uuid, gd.section_number,
           (gd.a_count*4 + gd.ab_count*3.5 + gd.b_count*3 + gd.bc_count*2.5
            + gd.c_count*2 + gd.d_count*1 + gd.f_count*0) AS pontos,
           (gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
            + gd.c_count + gd.d_count + gd.f_count) AS alunos
    FROM grade_distributions AS gd
    JOIN stat_offerings AS so ON gd.course_offering_uuid = so.course_offering_uuid
)
SELECT i.name AS professor,
       SUM(sg.pontos) AS soma_pontos,
       SUM(sg.alunos) AS total_alunos,
       SUM(sg.pontos) * 1.0 / SUM(sg.alunos) AS gpa_medio
FROM sec_gpa AS sg
JOIN sections AS se ON sg.course_offering_uuid = se.course_offering_uuid
                     AND sg.section_number = se.number
JOIN teachings AS t ON se.uuid = t.section_uuid
JOIN instructors AS i ON t.instructor_id = i.id
GROUP BY i.id
HAVING total_alunos > 0
ORDER BY gpa_medio
LIMIT 1

```

Table 3: 1 records

professor	soma_pontos	total_alunos	gpa_medio
JAMES D KUELBS	105.5	40	2.6375

- Quem é o professor mais fácil?

```

WITH stat_offerings AS (
    SELECT sm.course_offering_uuid
    FROM subject_memberships AS sm
    JOIN subjects AS su ON sm.subject_code = su.code
    WHERE su.abbreviation = 'STAT'
),
sec_gpa AS (
    SELECT gd.course_offering_uuid, gd.section_number,
           (gd.a_count*4 + gd.ab_count*3.5 + gd.b_count*3 + gd.bc_count*2.5
            + gd.c_count*2 + gd.d_count*1 + gd.f_count*0) AS pontos,

```

```

        (gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
        + gd.c_count + gd.d_count + gd.f_count) AS alunos
FROM grade_distributions AS gd
JOIN stat_offerings AS so ON gd.course_offering_uuid = so.course_offering_uuid
)
SELECT i.name AS professor,
       SUM(sg.pontos) AS soma_pontos,
       SUM(sg.alunos) AS total_alunos,
       SUM(sg.pontos) * 1.0 / SUM(sg.alunos) AS gpa_medio
FROM sec_gpa AS sg
JOIN sections AS se ON sg.course_offering_uuid = se.course_offering_uuid
                     AND sg.section_number = se.number
JOIN teachings AS t ON se.uuid = t.section_uuid
JOIN instructors AS i ON t.instructor_id = i.id
GROUP BY i.id
HAVING total_alunos > 0
ORDER BY gpa_medio DESC
LIMIT 1

```

Table 4: 1 records

professor	soma_pontos	total_alunos	gpa_medio
YAJUAN SI	44	11	4

- Qual é a disciplina mais difícil?

```

WITH stat_offerings AS (
  SELECT sm.course_offering_uuid
  FROM subject_memberships AS sm
  JOIN subjects AS su ON sm.subject_code = su.code
  WHERE su.abbreviation = 'STAT'
),
sec_gpa AS (
  SELECT gd.course_offering_uuid, gd.section_number,
         (gd.a_count*4 + gd.ab_count*3.5 + gd.b_count*3 + gd.bc_count*2.5
         + gd.c_count*2 + gd.d_count*1 + gd.f_count*0) AS pontos,
         (gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
         + gd.c_count + gd.d_count + gd.f_count) AS alunos
  FROM grade_distributions AS gd
  JOIN stat_offerings AS so ON gd.course_offering_uuid = so.course_offering_uuid
)

```

```

SELECT co.course_uuid,
       MIN(co.name) AS disciplina,
       SUM(sg.pontos) AS soma_pontos,
       SUM(sg.alunos) AS total_alunos,
       SUM(sg.pontos) * 1.0 / SUM(sg.alunos) AS gpa_medio
FROM sec_gpa AS sg
JOIN course_offerings AS co ON sg.course_offering_uuid = co.uuid
GROUP BY co.course_uuid
HAVING total_alunos > 0
ORDER BY gpa_medio
LIMIT 1

```

Table 5: 1 records

course_uuid	disciplina	soma_pontos	total_alunos	gpa_medio
3ef3ba60-426e-39ad-a61f-17d381b193d6	Intro-Theory of Probability	6439.5	2231	2.886374

- Qual é a disciplina mais fácil?

```

WITH stat_offerings AS (
  SELECT sm.course_offering_uuid
  FROM subject_memberships AS sm
  JOIN subjects AS su ON sm.subject_code = su.code
  WHERE su.abbreviation = 'STAT'
),
sec_gpa AS (
  SELECT gd.course_offering_uuid, gd.section_number,
         (gd.a_count*4 + gd.ab_count*3.5 + gd.b_count*3 + gd.bc_count*2.5
          + gd.c_count*2 + gd.d_count*1 + gd.f_count*0) AS pontos,
         (gd.a_count + gd.ab_count + gd.b_count + gd.bc_count
          + gd.c_count + gd.d_count + gd.f_count) AS alunos
  FROM grade_distributions AS gd
  JOIN stat_offerings AS so ON gd.course_offering_uuid = so.course_offering_uuid
)
SELECT co.course_uuid,
       MIN(co.name) AS disciplina,
       SUM(sg.pontos) AS soma_pontos,
       SUM(sg.alunos) AS total_alunos,
       SUM(sg.pontos) * 1.0 / SUM(sg.alunos) AS gpa_medio

```

```
FROM sec_gpa AS sg
JOIN course_offerings AS co ON sg.course_offering_uuid = co.uuid
GROUP BY co.course_uuid
HAVING total_alunos > 0
ORDER BY gpa_medio DESC
LIMIT 1
```

Table 6: 1 records

course_uuid	disciplina	soma_pontos	total_alunos	gpa_medio
99d2f325-3f56-397f-8c71-d4a2490a8e03	Empir Proc&Semiparmtrc Infernc	32	8	4

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-23 19:43:01 -03"
```